MASTER ROADMAP

Competitive Programming · DSA · DevOps

- Competitive Programming (16 Weeks)
- DevOps & Containerization (8 Weeks)
- Codeforces Specialist/Expert Target
- Interview Ready (LC Medium/Hard)
- Docker, Kubernetes, CI/CD
- 800+ Problems with Logic Building Focus

High-Intensity • Zero-Fluff • Results-Oriented

Total Problems: 800+

Total Duration: 16 + 8 Weeks

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Part I

Foundations & Training System

Chapter 1

Ground Rules, Philosophy & Training System

RULES YOU MUST FOLLOW — NO EXCEPTIONS

1. You solve problems **EVERY SINGLE DAY**. No zero days.
2. You participate in at **LEAST 2 live/virtual contests per week**.
3. You **UPSOLVE** every problem you could not finish in contest.
4. You maintain a **MISTAKE LOG** (Google Sheet / Notion).
5. You re-solve failed problems after **3 days**, then after **7 days**.
6. You **NEVER** copy-paste solutions. Type everything yourself.
7. You **TIME** yourself on every problem. Track solve times.
8. You **DRAW EXAMPLES** on paper before writing a single line of code.

1.1 Daily Time Commitment

Day Type	Hours	Activity
Weekday	3–4 hrs	1 hr theory + 2–3 hrs problem solving
Weekend	4–6 hrs	Contest + upsolving + revision

1.2 Problem-Solving Time Rules

Difficulty	Think Time	Action After
Easy	15 min	Check approach only (not code), then implement
Medium	30 min	Read editorial's first hint. Think 15 more min.
Hard	45 min	Read editorial. Code yourself. Re-solve in 3 days.

1.3 Logic Building Framework — The 5-Step Process

Since you struggle with logic building, internalize this for EVERY problem:

1. **REWRITE** the problem in your own words (2 min)
2. **DRAW** at least 3 examples: small, edge case, large (5 min)
3. **BRUTE FORCE** first — what is the $O(N^2)$ or $O(N^3)$ solution? (5 min)
4. **OPTIMIZE** — Can you sort it? Use hashmap? Binary search? DP? (10 min)
5. **CODE** only after you can explain the approach out loud (implement)

1.4 Weekly Schedule Template

Day	Activity	Duration
Mon	Learn new topic + solve 2 easy problems	3 hrs
Tue	Solve 3 medium problems on the topic	3 hrs
Wed	Solve 2 hard + 1 revision problem	3 hrs
Thu	Solve 3 mixed problems (prev + current)	3 hrs
Fri	Timed practice: 3 problems in 90 min	2+ hrs
Sat	**Live Contest** (CF/AtCoder) + upsolve	4–5 hrs
Sun	**Virtual Contest** + upsolve + mistake log	4–5 hrs

1.5 Progress Milestones

Week	Milestone	CF Rating
4	Comfortable solving Div2 A, B in < 15 min	1000–1200
8	Can solve Div2 C in $\sim$30 min, sometimes D	1200–1400
12	Consistently solve Div2 C, can attempt D/E	1400–1600
16	Solve Div2 D regularly, sometimes E	1600+

1.6 Mistake Log Format

Date	Problem	Why Failed	Correct Approach	Re-solve	

Chapter 2

Constraint → Complexity Mapping & Pattern Recognition

MEMORIZE THIS TABLE. Look at N first, ALWAYS.

Constraint	Expected Complexity
$N \leq 10$	$O(N!)$ brute force / bitmask
$N \leq 20$	$O(2^N)$ bitmask DP
$N \leq 100$	$O(N^3)$ Floyd / Interval DP
$N \leq 500$	$O(N^3)$ careful / $O(N^2 \log N)$
$N \leq 5000$	$O(N^2)$ DP / two pointers
$N \leq 10^5$	$O(N \log N)$ sorting / BS / seg tree
$N \leq 10^6$	$O(N)$ or $O(N \log N)$
$N \leq 10^9$	$O(\log N)$ binary search / math
$N \leq 10^{18}$	$O(\log N)$ binary exponentiation / digit DP

2.1 Pattern Recognition Checklist

Run through this checklist mentally for **every** problem:

- Asking for **min/max**? → DP, BS on answer, Greedy
- Asking for **count**? → DP, Combinatorics, PIE
- Asking for **existence**? → DP, Graph, 2-SAT
- **Order** matters? → Stack, Topological Sort
- **Graph** structure? → BFS/DFS/Dijkstra/DSU
- **Tree** structure? → DFS, LCA, Tree DP
- **Subset** problem? → Knapsack, Bitmask DP
- **Range query**? → Prefix sum, Seg Tree, BIT
- **Divisibility/primes**? → Number Theory
- **String pattern**? → KMP, Z, Trie, Hashing
- **Monotonic answer**? → Binary Search on Answer
- **XOR problem**? → Think bit-by-bit
- **Maximize minimum / Minimize maximum**? → BS on answer
- **Optimal substructure**? → DP
- **Exchange argument**? → Greedy
- **Connected components**? → DSU or BFS/DFS

Chapter 3

Essential Code Templates (C++)

3.1 Starter Template

```
#include <bits/stdc++.h>
using namespace std;
#define int long long
#define all(x) (x).begin(), (x).end()
#define sz(x) (int)(x).size()
#define pb push_back
#define pii pair<int,int>
#define vi vector<int>
const int MOD = 1e9 + 7;
const int INF = 1e18;

void solve() {
    int n;
    cin >> n;
    // YOUR CODE HERE
}

signed main() {
    ios_base::sync_with_stdio(false);
    cin.tie(NULL);
    int t = 1;
    cin >> t;
    while (t--) solve();
    return 0;
}
```

3.2 Binary Exponentiation

```
long long power(long long base, long long exp, long long mod) {
    long long result = 1;
    base %= mod;
    while (exp > 0) {
        if (exp & 1) result = result * base % mod;
```

```
        base = base * base % mod;
        exp >>= 1;
    }
    return result;
}
long long modInverse(long long a, long long mod) {
    return power(a, mod - 2, mod); // mod must be prime
}
```

3.3 Sieve of Eratosthenes

```
vector<bool> sieve(int n) {
    vector<bool> is_prime(n + 1, true);
    is_prime[0] = is_prime[1] = false;
    for (int i = 2; i * i <= n; i++)
        if (is_prime[i])
            for (int j = i * i; j <= n; j += i)
                is_prime[j] = false;
    return is_prime;
}
// Linear Sieve (O(N))
vector<int> smallest_prime(int n) {
    vector<int> sp(n + 1, 0), primes;
    for (int i = 2; i <= n; i++) {
        if (!sp[i]) { sp[i] = i; primes.push_back(i); }
        for (int p : primes) {
            if (p > sp[i] || (long long)i * p > n) break;
            sp[i * p] = p;
        }
    }
    return sp;
}
```

3.4 DSU (Union-Find)

```
struct DSU {
    vector<int> parent, rank_;
    int components;
    DSU(int n) : parent(n), rank_(n, 0), components(n) {
        iota(parent.begin(), parent.end(), 0);
    }
    int find(int x) {
        if (parent[x] != x) parent[x] = find(parent[x]);
        return parent[x];
    }
    bool unite(int x, int y) {
        x = find(x); y = find(y);
        if (x == y) return false;
        if (rank_[x] < rank_[y]) swap(x, y);
```

```
        parent[y] = x;
        if (rank_[x] == rank_[y]) rank_[x]++;
        components--;
        return true;
    }
    bool same(int x, int y) { return find(x) == find(y); }
};
```

3.5 Fenwick Tree (BIT)

```
struct BIT {
    int n;
    vector<long long> tree;
    BIT(int n) : n(n), tree(n + 1, 0) {}

    void update(int i, long long delta) {
        for (; i <= n; i += i & (-i))
            tree[i] += delta;
    }
    long long query(int i) {
        long long sum = 0;
        for (; i > 0; i -= i & (-i))
            sum += tree[i];
        return sum;
    }
    long long query(int l, int r) { return query(r) - query(l -
        1); }

    // Point update, range query BIT for minimum (different
        approach)
    // For range update point query: update(l, v), update(r+1, -v
        ), query(i)
};

// 2D BIT
struct BIT2D {
    int n, m;
    vector<vector<long long>> tree;
    BIT2D(int n, int m) : n(n), m(m), tree(n+1, vector<long long
        >(m+1,0)) {}
    void update(int x, int y, long long v) {
        for (int i = x; i <= n; i += i & -i)
            for (int j = y; j <= m; j += j & -j)
                tree[i][j] += v;
    }
    long long query(int x, int y) {
        long long s = 0;
        for (int i = x; i > 0; i -= i & -i)
            for (int j = y; j > 0; j -= j & -j)
                s += tree[i][j];
```

```
        return s;
    }
    long long query(int x1,int y1,int x2,int y2){
        return query(x2,y2)-query(x1-1,y2)-query(x2,y1-1)+query(
            x1-1,y1-1);
    }
};
```

3.6 Sparse Table (O(1) RMQ)

```
struct SparseTable {
    int n, LOG;
    vector<vector<int>> table;
    vector<int> log2_;

    SparseTable(vector<int>& arr) {
        n = arr.size();
        LOG = __lg(n) + 1;
        table.assign(LOG, vector<int>(n));
        log2_.resize(n + 1);
        log2_[1] = 0;
        for (int i = 2; i <= n; i++)
            log2_[i] = log2_[i / 2] + 1;
        table[0] = arr;
        for (int j = 1; j < LOG; j++)
            for (int i = 0; i + (1 << j) <= n; i++)
                table[j][i] = min(table[j-1][i],
                                  table[j-1][i + (1 << (j-1))]);
    }
    // O(1) range minimum query
    int query(int l, int r) {
        int k = log2_[r - l + 1];
        return min(table[k][l], table[k][r - (1 << k) + 1]);
    }
};
```

3.7 Segment Tree with Lazy Propagation

```
struct SegTree {
    int n;
    vector<long long> tree, lazy;
    SegTree(int n) : n(n), tree(4*n, 0), lazy(4*n, 0) {}

    void push(int v, int tl, int tr) {
        if (lazy[v]) {
            int tm = (tl + tr) / 2;
            apply(2*v, tl, tm, lazy[v]);
            apply(2*v+1, tm+1, tr, lazy[v]);
            lazy[v] = 0;
```

```
        }
    }
    void apply(int v, int tl, int tr, long long val) {
        tree[v] += val * (tr - tl + 1);
        lazy[v] += val;
    }
    void update(int v,int tl,int tr,int l,int r,long long val) {
        if (l > r) return;
        if (l == tl && r == tr) { apply(v, tl, tr, val); return;
           }
        push(v, tl, tr);
        int tm = (tl + tr) / 2;
        update(2*v, tl, tm, l, min(r, tm), val);
        update(2*v+1, tm+1, tr, max(l, tm+1), r, val);
        tree[v] = tree[2*v] + tree[2*v+1];
    }
    long long query(int v, int tl, int tr, int l, int r) {
        if (l > r) return 0;
        if (l == tl && r == tr) return tree[v];
        push(v, tl, tr);
        int tm = (tl + tr) / 2;
        return query(2*v,tl,tm,l,min(r,tm))
             + query(2*v+1,tm+1,tr,max(l,tm+1),r);
    }
};
```

3.8 LCA — Binary Lifting

```
const int LOG = 20;
int up[200005][LOG], depth[200005];

void dfs(int v, int p, vector<vector<int>>& adj) {
    up[v][0] = p;
    for (int i = 1; i < LOG; i++)
        up[v][i] = up[up[v][i-1]][i-1];
    for (int u : adj[v]) if (u != p) {
        depth[u] = depth[v] + 1;
        dfs(u, v, adj);
    }
}
int lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    int diff = depth[u] - depth[v];
    for (int i = 0; i < LOG; i++)
        if ((diff >> i) & 1) u = up[u][i];
    if (u == v) return u;
    for (int i = LOG - 1; i >= 0; i--)
        if (up[u][i] != up[v][i]) {
            u = up[u][i]; v = up[v][i];
        }
```

```
    return up[u][0];
}
```

3.9 Combinatorics (nCr mod p)

```
const int MAXN = 200005;
const int MOD = 1e9 + 7;
long long fact[MAXN], inv_fact[MAXN];

void precompute() {
    fact[0] = 1;
    for (int i = 1; i < MAXN; i++)
        fact[i] = fact[i-1] * i % MOD;
    inv_fact[MAXN-1] = power(fact[MAXN-1], MOD-2, MOD);
    for (int i = MAXN-2; i >= 0; i--)
        inv_fact[i] = inv_fact[i+1] * (i+1) % MOD;
}
long long C(int n, int r) {
    if (r < 0 || r > n) return 0;
    return fact[n] % MOD * inv_fact[r] % MOD * inv_fact[n-r] %
       MOD;
}
```

Part II

The 800+ Problem Master List

Chapter 4

Two Pointers & Sliding Window

GOAL

Master these 3 patterns first. They appear in 40%+ of contest problems in some form. **Two Pointers** → Sorted array, opposite ends or fast-slow. **Sliding Window** → Contiguous subarray/substring problems. **Prefix Sum** → Range sum/count queries in $O(1)$.

#	Problem	Platform	Diff	Key Pattern
1	Two Sum II (167)	LC	Easy	Opposite ends
2	Two Sum (1)	LC	Easy	Hashmap
3	3Sum (15)	LC	Medium	Fix one + two-pointer
4	3Sum Closest (16)	LC	Medium	Same as 3Sum
5	4Sum (18)	LC	Medium	Nested two pointers
6	4Sum II (454)	LC	Medium	Two hashmaps
7	Container With Most Water (11)	LC	Medium	Greedy two-pointer
8	Trapping Rain Water (42)	LC	Hard	Two-pointer / stack
9	Remove Duplicates from Sorted	LC	Easy	Fast-slow pointer
10	Valid Palindrome (125)	LC	Easy	Two-pointer from ends
11	Longest Substr Without Repeat	LC	Medium	Window + set
12	Minimum Window Substring (76)	LC	Hard	Template problem
13	Permutation in String (567)	LC	Medium	Fixed-size window
14	Find All Anagrams (438)	LC	Medium	Sliding window
15	Longest Rep Char Replacement	LC	Medium	Window + frequency
16	Max Consecutive Ones III	LC	Medium	At-most-K zeros
17	Subarray Product Less Than K	LC	Medium	Shrinking window
18	Minimum Size Subarray Sum	LC	Medium	Expand-contract
19	Fruit Into Baskets (904)	LC	Medium	At-most 2 distinct

#	Problem	Platform	Diff	Key Pattern
20	Subarrays K Different Int	LC	Hard	At-most trick
21	Longest Subarray of 1s (1493)	LC	Medium	Sliding window
22	Max Erasure Value (1695)	LC	Medium	Distinct window
23	Minimum Operations to Reduce X	LC	Medium	Complement + window
24	Get Equal Substrings (1208)	LC	Medium	Sliding window
25	Count Vowel Substrings (2062)	LC	Medium	At-most trick
26	Sum of Two Values	CSES	Easy	Sorted + two-ptr
27	Subarray Sums I	CSES	Easy	Prefix + hashmap
28	Subarray Sums II	CSES	Medium	Prefix + map
29	Subarray Divisibility	CSES	Medium	Modulo trick
30	Subarray Distinct Values	CSES	Medium	Sliding window
31	Books (CF 279B)	CF	Medium	Classic sliding
32	Number of Equal Substr (1296C)	CF	Medium	Two-pointer
33	Sereja and Dima (CF 381A)	CF	Easy	Two-pointer game
34	Polycarp Training (CF 1165B)	CF	Easy	Greedy + sort
35	Infinite Sequence (CF 1216C)	CF	Medium	Sliding window
36	Vasya and String (CF 1025C)	CF	Medium	Sliding window
37	Maximum Sum (CF 1303C)	CF	Medium	Prefix + greedy
38	Prefix Sums (CF 1476C)	CF	Medium	Prefix logic
39	Omkar and the Len. Game (1586)	CF	Easy	Observation
40	Good Subarrays (CF 1398C)	CF	Medium	Prefix hash
41	Two Buttons (CF 520B)	CF	Medium	BFS / two-ptr
42	Say No to Palindromes (1555D)	CF	Hard	Prefix + brute
43	Approximating Constant Range	CF	Hard	Sliding window
44	Nastya and Doors (CF 1382B)	CF	Medium	Observation
45	Playlist	CSES	Medium	Sliding window
46	Forest Queries	CSES	Medium	2D prefix sum
47	Range Queries and Copies	CSES	Hard	Persistent DS

Chapter 5

Binary Search

KEY INSIGHT: If you can write a function `check(x)` that returns True/False, and the answers form `[True,True,...,True,False,False,...]`, then binary search on `x`.

#	Problem	Platform	Diff	Key Pattern
1	Binary Search (704)	LC	Easy	Template
2	Search Insert Position (35)	LC	Easy	Lower bound
3	First Bad Version (278)	LC	Easy	BS on bool
4	Find Smallest Letter (744)	LC	Easy	Circular BS
5	Peak Index in Mountain (852)	LC	Easy	Peak finding
6	Guess Number Higher or Lower	LC	Easy	Classic BS
7	Find First and Last Position (34)	LC	Medium	Lower/upper bound
8	Find Peak Element (162)	LC	Medium	Local peak
9	Search in Rotated Array (33)	LC	Medium	Modified BS
10	Search in Rotated II (81)	LC	Medium	With duplicates
11	Find Min in Rotated (153)	LC	Medium	Min finding
12	Find Min in Rotated II (154)	LC	Hard	With duplicates
13	Search a 2D Matrix (74)	LC	Medium	Flatten + BS
14	Search a 2D Matrix II (240)	LC	Medium	Staircase search
15	Median of Two Sorted Arrays (4)	LC	Hard	Partition BS
16	Koko Eating Bananas (875)	LC	Medium	BS on answer
17	Capacity to Ship Packages (1011)	LC	Medium	BS on capacity
18	Split Array Largest Sum (410)	LC	Hard	BS on max sum
19	Magnetic Force Between Balls	LC	Medium	BS on distance
20	Minimize Max Distance to Gas	LC	Hard	BS on answer
21	Kth Smallest in Mult Table (668)	LC	Hard	BS + math
22	Find K-th Smallest Pair Dist	LC	Hard	BS + two-ptr

#	Problem	Platform	Diff	Key Pattern
23	K-th Smallest Prime Fraction	LC	Hard	BS on fraction
24	Swim in Rising Water (778)	LC	Hard	BS + BFS
25	Nth Root of Integer	GFG	Medium	BS on answer
26	Aggressive Cows	SPOJ	Medium	Classic spacing
27	PRATA (Parathas)	SPOJ	Medium	BS on time
28	EKO (Eko)	SPOJ	Easy	BS on height
29	Ropes	SPOJ	Medium	BS on length
30	SCALE (BS)	SPOJ	Medium	BS + greedy
31	Factory Machines	CSES	Medium	BS on time
32	Array Division	CSES	Medium	BS on max sub
33	Maximum Median (CF 1201C)	CF	Medium	BS on median
34	Letters (CF 1187B)	CF	Medium	BS on prefix
35	Balanced Stone Heaps (CF 1623C)	CF	Hard	BS on answer
36	Poisoned Dagger (CF 1613C)	CF	Medium	BS on time
37	Devu and Flowers (CF 451C)	CF	Hard	Stars + bars + IE
38	Ants Colony (CF 704D)	CF	Hard	BS + LCA
39	Binary Search (CF 1223C)	CF	Medium	BS on index
40	Infinite Inversions (CF 1744E)	CF	Hard	BS + math
41	Vacations (CF 1472G)	CF	Hard	BS on answer
42	Minimize the Diameter (CF 1854C)	CF	Hard	BS + tree
43	B. Kira and the Golden Rule	CF	Medium	Observation
44	Minimizing the Difference (1473C)	CF	Medium	BS + greedy
45	Wood Cutting (CSES)	CSES	Medium	BS on height

Chapter 6

Bit Manipulation & Number Theory

6.1 Bit Manipulation

#	Problem	Platform	Diff	Pattern
1	Single Number (136)	LC	Easy	XOR
2	Single Number II (137)	LC	Medium	Bit count mod 3
3	Single Number III (260)	LC	Medium	XOR + bit split
4	Missing Number (268)	LC	Easy	XOR
5	Number of 1 Bits (191)	LC	Easy	Brian Kernighan
6	Counting Bits (338)	LC	Easy	DP on bits
7	Reverse Bits (190)	LC	Easy	Bit ops
8	Power of Two (231)	LC	Easy	n & (n-1)
9	Power of Four (342)	LC	Easy	Bit pattern
10	Bitwise AND of Range (201)	LC	Medium	Common prefix
11	Maximum XOR of Two Numbers (421)	LC	Medium	Trie
12	Sum of Two Integers (371)	LC	Medium	Bit adder
13	Subsets (78)	LC	Medium	Bitmask gen
14	Subsets II (90)	LC	Medium	Bitmask + dedup
15	XOR Queries of Subarray (1310)	LC	Medium	Prefix XOR
16	Decode XORed Array (1720)	LC	Easy	XOR property
17	Count Triplets with XOR=0 (1442)	LC	Medium	Nested XOR
18	Maximum XOR With Element (1707)	LC	Hard	Trie + offline
19	Divide Two Integers (29)	LC	Medium	Bit division
20	Reverse Integer (7)	LC	Medium	Bit/math
21	And Then There Were K (CF 1527A)	CF	Easy	Observation
22	Gardener and Array (CF 1775B)	CF	Medium	Bit ops
23	Mike and Frog (CF 547A)	CF	Hard	Cycle detection
24	The Child and Toy (CF 437B)	CF	Easy	Greedy + bit

#	Problem	Platform	Diff	Pattern
25	Numbers (CF 1265B)	CF	Easy	XOR observation
26	XOR-ranges (CF 1650E)	CF	Hard	Bit DP
27	Kth Bit in Nth Binary String	LC	Medium	Recursion
28	Bits	CSES	Medium	Bit logic
29	Reachable Numbers	CSES	Medium	Graph + bit
30	XOR Pyramid (CF 1547F)	CF	Hard	Bit + DP

6.2 Number Theory

#	Problem	Platform	Diff	Pattern
1	GCD of Strings (1071)	LC	Easy	GCD
2	Count Primes (204)	LC	Medium	Sieve
3	Ugly Number II (264)	LC	Medium	Min-heap / DP
4	Super Ugly Number (313)	LC	Medium	Heap + prime
5	Nth Digit (400)	LC	Medium	Math
6	Super Pow (372)	LC	Medium	Mod exp
7	Pow(x, n) (50)	LC	Medium	Fast exp
8	Nth Magical Number (878)	LC	Hard	BS + LCM
9	GCD Sort (1998)	LC	Hard	DSU + prime
10	Exponentiation	CSES	Easy	Basic mod exp
11	Exponentiation II	CSES	Medium	Euler's theorem
12	Counting Divisors	CSES	Medium	Factorize all
13	Common Divisors	CSES	Medium	GCD + sieve
14	Sum of Divisors	CSES	Hard	Harmonic series
15	Prime Multiples	CSES	Hard	Inclusion-Exclusion
16	Goldbach's Conjecture	CSES	Hard	Sieve
17	Fibonacci Numbers	CSES	Medium	Matrix exp
18	Throwing Dice	CSES	Medium	Matrix exp
19	Graph Paths I	CSES	Medium	Matrix exp
20	Graph Paths II	CSES	Hard	Matrix exp
21	Divisor Analysis	CSES	Medium	Factorization
22	Candy Division	CSES	Easy	GCD
23	Binomial Coefficients	CSES	Easy	nCr mod p
24	Creating Strings	CSES	Easy	Permutations
25	Distributing Apples	CSES	Medium	Stars and bars
26	Christmas Party	CSES	Medium	Derangements
27	Bracket Sequences I	CSES	Hard	Catalan numbers
28	Bracket Sequences II	CSES	Hard	DP + Catalan
29	Product of Three Numbers (1294C)	CF	Medium	Factorization
30	GCD and MST (CF 1513D)	CF	Hard	DSU + math
31	Wet Shark and Flowers (CF 621C)	CF	Medium	GCD observation

#	Problem	Platform	Diff	Pattern
32	New Year and the Olympiad (750D)	CF	Hard	Math + DP
33	Vasya and a Tree (CF 1017E)	CF	Hard	DFS + BIT + math
34	Anton and Diofant (CF 632D)	CF	Medium	GCD
35	Almost Arithmetic (CF 1796C)	CF	Medium	Math

Chapter 7

Greedy Algorithms

#	Problem	Platform	Diff	Pattern
1	Assign Cookies (455)	LC	Easy	Sort + greedy
2	Lemonade Change (860)	LC	Easy	Simulation
3	Queue Reconstr by Height (406)	LC	Medium	Sort twice
4	Non-overlapping Intervals (435)	LC	Medium	Interval sched
5	Minimum Arrows to Burst (452)	LC	Medium	Interval merge
6	Partition Labels (763)	LC	Medium	Last occurrence
7	Jump Game (55)	LC	Medium	Greedy reach
8	Jump Game II (45)	LC	Medium	BFS-like
9	Gas Station (134)	LC	Medium	Circular greedy
10	Task Scheduler (621)	LC	Medium	Max frequency
11	Reorganize String (767)	LC	Medium	Greedy + heap
12	Candy (135)	LC	Hard	Two-pass
13	Boats to Save People (881)	LC	Medium	Two-pointer
14	Advantage Shuffle (870)	LC	Medium	Sort + match
15	Minimum Platforms	GFG	Medium	Events + sort
16	Activity Selection	GFG	Easy	Classic greedy
17	Fractional Knapsack	GFG	Easy	Sort by ratio
18	Movie Festival	CSES	Easy	Interval sched
19	Movie Festival II	CSES	Medium	Multi-resource
20	Maximum Subarray Sum	CSES	Easy	Kadane's
21	Stick Lengths	CSES	Easy	Median
22	Missing Coin Sum	CSES	Medium	Greedy prefix
23	Collecting Numbers	CSES	Medium	Greedy sweep
24	Collecting Numbers II	CSES	Hard	Seg tree variant
25	Tasks and Deadlines	CSES	Medium	Sort by deadline
26	Reading Books	CSES	Medium	Max element
27	Sum of Three Values	CSES	Medium	Greedy + hash
28	Towers	CSES	Medium	Multiset greedy
29	Traffic Lights	CSES	Medium	Set operations

#	Problem	Platform	Diff	Pattern
30	Nested Ranges Check	CSES	Medium	Sorting + stack
31	Nested Ranges Count	CSES	Hard	Coord compress
32	Constructing the Array (CF 1353D)	CF	Medium	Greedy + heap
33	The Sports Festival (CF 1509C)	CF	Hard	BS + greedy
34	Increasing Subsequence (1157C2)	CF	Hard	Greedy
35	Greedy Subsequences (CF 1765E)	CF	Hard	Greedy + stack
36	Monsters (CF 593B)	CF	Medium	Greedy sort
37	Queue Sorting (CF 1561C)	CF	Easy	Observation
38	Minimize the Thickness (CF 1838C)	CF	Easy	Greedy
39	Making It Beautiful (CF 1264C)	CF	Medium	Greedy
40	Vanya and Lanterns (CF 492C)	CF	Easy	Greedy
41	Make It Good (CF 1385C)	CF	Easy	Observation
42	Polycarp and Sums (CF 1732C)	CF	Medium	Greedy
43	Three-pile Nim (CF 768C)	CF	Easy	Observation
44	Balanced Rating Changes (CF 1503B)	CF	Medium	Greedy
45	Salary Increasing (CF 1550C)	CF	Medium	Greedy + BS

Chapter 8

Stack, Queue & Heap

#	Problem	Platform	Diff	Pattern
1	Next Greater Element I (496)	LC	Easy	Monotonic stack
2	Next Greater Element II (503)	LC	Medium	Circular stack
3	Daily Temperatures (739)	LC	Medium	Mono stack
4	Online Stock Span (901)	LC	Medium	Mono stack
5	Largest Rect in Histogram (84)	LC	Hard	Mono stack
6	Maximal Rectangle (85)	LC	Hard	Hist per row
7	Trapping Rain Water (42)	LC	Hard	Stack / two-ptr
8	Sum of Subarray Minimums (907)	LC	Medium	Mono stack
9	Sum of Subarray Ranges (2104)	LC	Medium	Mono stack
10	132 Pattern (456)	LC	Medium	Mono stack
11	Remove K Digits (402)	LC	Medium	Mono stack
12	Remove Duplicate Letters (316)	LC	Medium	Mono stack
13	Largest Rectangle in Matrix	GFG	Hard	Hist extension
14	Maximum Width Ramp (962)	LC	Medium	Mono stack
15	Sliding Window Maximum (239)	LC	Hard	Mono deque
16	Shortest Subarray Sum $\geq$ K	LC	Hard	Mono deque
17	Jump Game VI (1696)	LC	Medium	Deque + DP
18	Constrained Subseq Sum (1425)	LC	Hard	Deque + DP
19	Min Stack (155)	LC	Medium	Design
20	LRU Cache (146)	LC	Medium	HashMap + DLL
21	LFU Cache (460)	LC	Hard	Design
22	Kth Largest Element (215)	LC	Medium	Heap / quicksel
23	Find Median from Stream (295)	LC	Hard	Two heaps
24	Sliding Window Median (480)	LC	Hard	Two heaps

#	Problem	Platform	Diff	Pattern
25	Top K Frequent Elements (347)	LC	Medium	Heap / bucket
26	Merge K Sorted Lists (23)	LC	Hard	Min-heap
27	K Pairs with Smallest Sums (373)	LC	Medium	Heap
28	K Closest Points to Origin (973)	LC	Medium	Heap
29	Task Scheduler (621)	LC	Medium	Max-heap
30	Reorganize String (767)	LC	Medium	Max-heap
31	IPO (502)	LC	Hard	Two heaps
32	The Skyline Problem (218)	LC	Hard	Heap + events
33	Nearest Smaller Values	CSES	Easy	Mono stack
34	Advertisement	CSES	Medium	Histogram
35	Subarray Squares	CSES	Hard	Mono deque
36	Largest Building (CSES)	CSES	Hard	Histogram
37	Max Score from Removals (CF 1737D)	CF	Hard	Heap + greedy
38	Monoblock (CF 1528D)	CF	Hard	Mono stack
39	Bracket Sequences (CF 380C)	CF	Medium	Stack
40	Stack Sorting (CF 1439C)	CF	Medium	Stack simulation
41	Maximum in Sliding Window (CF)	CF	Medium	Mono deque
42	Number of Pairs (CF 1538C)	CF	Medium	Mono stack
43	Simple Polygon (CF 1040C)	CF	Medium	Stack
44	Sequence of Containers (CF 1796D)	CF	Hard	Priority queue
45	Playlist	CSES	Medium	Sliding window

Chapter 9

Dynamic Programming

DP PROBLEM-SOLVING TEMPLATE (use every time):

1. **Define state:** What do I need to remember? `dp[i][j] = ???`
2. **Transition:** How do I get `dp[i][j]` from smaller states?
3. **Base case:** What is `dp[0][0]` or `dp[start]`?
4. **Answer:** Which dp value is the final answer?
5. **Order:** In what order do I fill the table?
6. **Optimization:** Can I reduce space? Use binary search (LIS)?

9.1 1D DP

#	Problem	Platform	Diff	Pattern
1	Climbing Stairs (70)	LC	Easy	Fibonacci DP
2	Min Cost Climbing Stairs (746)	LC	Easy	1D DP
3	House Robber (198)	LC	Medium	No-adjacent
4	House Robber II (213)	LC	Medium	Circular
5	Delete and Earn (740)	LC	Medium	Reduce to robber
6	Maximum Subarray (53)	LC	Medium	Kadane's
7	Maximum Product Subarray (152)	LC	Medium	Min/max tracking
8	Decode Ways (91)	LC	Medium	String DP
9	Decode Ways II (639)	LC	Hard	String DP
10	Coin Change (322)	LC	Medium	Unbounded KS
11	Coin Change II (518)	LC	Medium	Count combos
12	Combination Sum IV (377)	LC	Medium	Ordered combos
13	Perfect Squares (279)	LC	Medium	BFS / DP
14	Frog Jump (403)	LC	Hard	DP on set
15	Integer Break (343)	LC	Medium	Max product
16	Dice Combinations	CSES	Easy	1D DP
17	Minimizing Coins	CSES	Easy	Unbounded KS
18	Coin Combinations I	CSES	Easy	Ordered combos

#	Problem	Platform	Diff	Pattern
19	Coin Combinations II	CSES	Easy	Unordered combos
20	Removing Digits	CSES	Easy	BFS / DP
21	Array Description	CSES	Medium	State DP
22	Counting Towers	CSES	Medium	1D DP
23	Sereja and Brackets (CF 380C)	CF	Medium	Stack / DP
24	Arithmetic Progressions (CF 1826E)	CF	Hard	DP
25	Tanya and Postcard (CF 1359D)	CF	Medium	Greedy + DP

9.2 2D DP

#	Problem	Platform	Diff	Pattern
1	Unique Paths (62)	LC	Medium	Grid DP
2	Unique Paths II (63)	LC	Medium	Grid with obstacles
3	Minimum Path Sum (64)	LC	Medium	Grid DP
4	Triangle (120)	LC	Medium	Top-down / bottom
5	Dungeon Game (174)	LC	Hard	Reverse grid DP
6	Maximal Square (221)	LC	Medium	2D DP
7	Maximal Rectangle (85)	LC	Hard	Histogram per row
8	Cherry Pickup (741)	LC	Hard	Two-path DP
9	Cherry Pickup II (1463)	LC	Hard	Two-path grid
10	Grid Paths	CSES	Medium	Grid DP
11	Rectangle Cutting	CSES	Medium	Interval DP
12	Longest Flight Route	CSES	Medium	DAG DP
13	Counting Tilings	CSES	Hard	Bitmask DP
14	Counting Numbers	CSES	Hard	Digit DP

9.3 Knapsack DP

#	Problem	Platform	Diff	Pattern
1	Partition Equal Subset Sum (416)	LC	Medium	0/1 KS
2	Target Sum (494)	LC	Medium	0/1 KS
3	Last Stone Weight II (1049)	LC	Medium	0/1 KS
4	Ones and Zeroes (474)	LC	Medium	2D KS
5	Partition to K Equal Subsets	LC	Medium	Bitmask / backtrack
6	Rod Cutting	GFG	Medium	Unbounded KS
7	Book Shop	CSES	Medium	0/1 KS
8	Money Sums	CSES	Medium	0/1 KS count
9	Two Sets II	CSES	Medium	KS divisibility

#	Problem	Platform	Diff	Pattern
10	Projects	CSES	Hard	KS + sorting
11	Knapsack (CF 1030E)	CF	Hard	Subset sum
12	Vasya and Multisets (CF 1051E)	CF	Hard	DP optimization
13	Mike and Fun (CF 547B)	CF	Medium	Row-by-row DP
14	Divisible Subset (CF 1444B)	CF	Medium	KS variant
15	Ball in a Hole (CF 1700D)	CF	Hard	Knapsack

9.4 LIS / LCS

#	Problem	Platform	Diff	Pattern
1	Longest Increasing Subsequence	LC	Medium	DP / patience sort
2	Russian Doll Envelopes (354)	LC	Hard	LIS in 2D
3	Max Length of Pair Chain (646)	LC	Medium	Greedy LIS
4	Number of LIS (673)	LC	Medium	Count LIS
5	Longest Common Subsequence	LC	Medium	2D DP
6	Uncrossed Lines (1035)	LC	Medium	Reduce to LCS
7	Edit Distance (72)	LC	Medium	2D DP
8	Delete Op for Two Strings (583)	LC	Medium	LCS variant
9	Min ASCII Delete Sum (712)	LC	Medium	LCS with weights
10	Distinct Subsequences (115)	LC	Hard	Count subseq
11	Shortest Common Superseq (1092)	LC	Hard	LCS + reconstruct
12	Interleaving String (97)	LC	Medium	2D bool DP
13	Increasing Subsequence	CSES	Medium	LIS reconstruction
14	Edit Distance (CSES)	CSES	Medium	Classic
15	LIS (CF 1398D)	CF	Hard	Greedy LIS
16	Subsequence Counting (CF 1748E)	CF	Hard	DP + bitmask
17	Colorful Graph (CF 1891E)	CF	Hard	LCS variant

9.5 Interval / Matrix Chain DP

#	Problem	Platform	Diff	Pattern
1	Burst Balloons (312)	LC	Hard	Interval DP
2	Min Cost to Merge Stones (1000)	LC	Hard	Interval DP
3	Strange Printer (664)	LC	Hard	Interval DP

#	Problem	Platform	Diff	Pattern
4	Palindrome Partitioning II (132)	LC	Hard	Interval DP
5	Palindrome Partitioning III (1278)	LC	Hard	Interval DP
6	Min Cost to Cut a Stick (1547)	LC	Hard	Interval DP
7	Matrix Chain Multiplication	GFG	Hard	Classic MCM
8	Optimal BST	GFG	Hard	Interval DP
9	Zuma Game (CF 41D)	CF	Hard	Interval DP
10	Codeforces Problem (CF 1527E)	CF	Hard	Interval DP
11	Rectangle Cutting (CSES)	CSES	Medium	Interval DP
12	Removal Game (CSES)	CSES	Medium	Interval DP

9.6 Bitmask DP

#	Problem	Platform	Diff	Pattern
1	Shortest Path Visiting All Nodes	LC	Hard	BFS + bitmask
2	Find Shortest Superstring (943)	LC	Hard	TSP variant
3	Smallest Sufficient Team (1125)	LC	Hard	Bitmask KS
4	Stickers to Spell Word (691)	LC	Hard	Bitmask DP
5	Min Cost to Connect Groups (1595)	LC	Hard	Bitmask DP
6	Hamiltonian Flights	CSES	Medium	TSP
7	Elevator Rides	CSES	Hard	Bitmask DP
8	Counting Tilings	CSES	Hard	Profile bitmask
9	Kefa and Dishes (CF 580D)	CF	Hard	Bitmask DP
10	Tourist's Notes (CF 57C)	CF	Hard	Bitmask
11	Sherlock and Permutations (1696E)	CF	Hard	Bitmask
12	Maximum Subset (CF 1036D)	CF	Hard	Bitmask + DP

9.7 Digit DP

#	Problem	Platform	Diff	Pattern
1	Count Num with Unique Digits (357)	LC	Medium	Digit DP basics
2	Numbers At Most N (902)	LC	Hard	Digit DP
3	Number of Digit One (233)	LC	Hard	Digit DP / math

#	Problem	Platform	Diff	Pattern
4	Digit Count in Range	GFG	Hard	Digit DP
5	Counting Numbers	CSES	Hard	Digit DP
6	Classy Numbers (CF 1036C)	CF	Hard	Digit DP
7	Lucky Numbers (CF 95C)	CF	Medium	Digit DP
8	Magic Numbers (CF 628D)	CF	Medium	Digit DP
9	Password (CF 126B)	CF	Medium	KMP + digit DP
10	Palindromes (CF 906E)	CF	Hard	Digit + palindrome

9.8 DP on Trees

#	Problem	Platform	Diff	Pattern
1	House Robber III (337)	LC	Medium	Choose/skip
2	Longest Univalue Path (687)	LC	Medium	Tree DFS
3	Binary Tree Cameras (968)	LC	Hard	3-state DP
4	Sum of Distances in Tree (834)	LC	Hard	Rerooting
5	Tree Matching	CSES	Medium	2-state DP
6	Tree Distances I	CSES	Medium	DFS
7	Tree Distances II	CSES	Hard	Rerooting
8	Maximum White Subtree (CF 1324F)	CF	Hard	Rerooting DP
9	Tree with Max Cost (CF 1092F)	CF	Hard	Rerooting DP
10	Appleman and Tree (CF 461B)	CF	Medium	Tree DP
11	New Year Tree (CF 750E)	CF	Hard	Tree DP + seg
12	Xenia and Tree (CF 342E)	CF	Hard	Centroid + DP

Chapter 10

Trees

#	Problem	Platform	Diff	Pattern
1	Maximum Depth of BT (104)	LC	Easy	DFS
2	Minimum Depth of BT (111)	LC	Easy	BFS
3	Invert Binary Tree (226)	LC	Easy	DFS
4	Symmetric Tree (101)	LC	Easy	Mirror DFS
5	Same Tree (100)	LC	Easy	DFS
6	Balanced Binary Tree (110)	LC	Easy	Height DFS
7	Diameter of Binary Tree (543)	LC	Easy	DFS
8	Path Sum (112)	LC	Easy	DFS
9	Path Sum II (113)	LC	Medium	Backtracking
10	Path Sum III (437)	LC	Medium	Prefix sum
11	BT Maximum Path Sum (124)	LC	Hard	DFS
12	BT Level Order Traversal (102)	LC	Medium	BFS
13	BT Zigzag Level Order (103)	LC	Medium	BFS
14	BT Right Side View (199)	LC	Medium	BFS
15	Vertical Order Traversal (987)	LC	Hard	BFS + sort
16	Validate BST (98)	LC	Medium	Inorder
17	Kth Smallest in BST (230)	LC	Medium	Inorder
18	Lowest Common Ancestor (236)	LC	Medium	DFS
19	LCA of BST (235)	LC	Medium	BST property
20	All Nodes Distance K (863)	LC	Medium	BFS from node
21	Serialize/Deserialize BT (297)	LC	Hard	DFS encoding
22	Construct BT Pre/In (105)	LC	Medium	Divide & conquer
23	Construct BT In/Post (106)	LC	Medium	Divide & conquer
24	Flatten BT to Linked List (114)	LC	Medium	Morris/DFS
25	Recover BST (99)	LC	Medium	Inorder swap
26	Unique BSTs (96)	LC	Medium	Catalan / DP
27	House Robber III (337)	LC	Medium	Tree DP
28	Binary Tree Cameras (968)	LC	Hard	Tree DP
29	Sum of Distances in Tree (834)	LC	Hard	Rerooting

#	Problem	Platform	Diff	Pattern
30	Subordinates	CSES	Easy	DFS subtree
31	Tree Diameter	CSES	Medium	Double DFS
32	Tree Distances I	CSES	Medium	DFS
33	Tree Distances II	CSES	Hard	Rerooting
34	Company Queries I	CSES	Medium	Binary lifting
35	Company Queries II	CSES	Medium	LCA
36	Distance Queries	CSES	Medium	LCA
37	Counting Paths	CSES	Hard	LCA + diff
38	Subtree Queries	CSES	Medium	Euler tour + BIT
39	Path Queries	CSES	Medium	Euler tour + BIT
40	Path Queries II	CSES	Hard	HLD
41	Distinct Colors	CSES	Hard	DSU on tree
42	Fools and Roads (CF 191C)	CF	Hard	LCA + diff
43	Lomsat Gelral (CF 600E)	CF	Hard	DSU on tree
44	Tree Painting (CF 1187E)	CF	Hard	Tree DP + DFS
45	Maximum White Subtree (CF 1324F)	CF	Hard	Rerooting
46	New Year Tree (CF 750E)	CF	Hard	HLD + seg
47	Xenia and Tree (CF 342E)	CF	Hard	Centroid decomp
48	Vertex Add Path Sum (CF)	CF	Medium	HLD
49	Appleman and Tree (CF 461B)	CF	Medium	Tree DP
50	Fixed-Length Paths I	CSES	Hard	Centroid decomp

Chapter 11

Graphs

11.1 BFS / DFS

#	Problem	Platform	Diff	Pattern
1	Number of Islands (200)	LC	Medium	BFS/DFS flood
2	Max Area of Island (695)	LC	Medium	DFS
3	Flood Fill (733)	LC	Easy	DFS
4	Surrounded Regions (130)	LC	Medium	Boundary DFS
5	Pacific Atlantic Water Flow (417)	LC	Medium	Multi-source BFS
6	Rotting Oranges (994)	LC	Medium	Multi-source BFS
7	Walls and Gates (286)	LC	Medium	Multi-source BFS
8	Clone Graph (133)	LC	Medium	BFS + hashmap
9	Keys and Rooms (841)	LC	Medium	DFS
10	Find If Path Exists (1971)	LC	Easy	BFS / DSU
11	Labyrinth	CSES	Easy	BFS
12	Message Route	CSES	Easy	BFS path
13	Building Roads	CSES	Easy	Components BFS
14	Round Trip	CSES	Medium	DFS cycle
15	Building Teams	CSES	Medium	Bipartite
16	Download Speed	CSES	Medium	Max flow
17	School Dance	CSES	Hard	Bipartite matching
18	Distinct Routes	CSES	Hard	Disjoint paths
19	Solve The Maze (CF 1365D)	CF	Medium	BFS + construction
20	Codeforces (CF 1144F)	CF	Medium	Bipartite check
21	Police Stations (CF 796D)	CF	Medium	Multi-source BFS
22	Garland (CF 1491H)	CF	Hard	BFS layers
23	Water Tree (CF 343D)	CF	Hard	DFS + BIT
24	Reachable Nodes (CF 1234D)	CF	Medium	BFS reach
25	Checkered Pattern (CF 1527D)	CF	Medium	BFS coloring

11.2 Topological Sort

#	Problem	Platform	Diff	Pattern
1	Course Schedule (207)	LC	Medium	Detect cycle
2	Course Schedule II (210)	LC	Medium	Reconstruct order
3	Alien Dictionary (269)	LC	Hard	Topo sort
4	Sequence Reconstruction (444)	LC	Medium	Unique topo sort
5	Parallel Courses (1136)	LC	Medium	BFS levels
6	Find Eventual Safe States (802)	LC	Medium	Reverse graph
7	Course Schedule (CSES)	CSES	Medium	Classic
8	Longest Flight Route	CSES	Medium	DAG longest path
9	Game Routes	CSES	Medium	DAG count paths
10	Fox And Names (CF 510C)	CF	Medium	Topo sort
11	Minimal String (CF 797F)	CF	Hard	Topo + Euler path
12	Constract the String (CF 1556E)	CF	Hard	Topo DP
13	Timeline (CF 1448D)	CF	Medium	Topo sort
14	Work for Robots (CF 1703F)	CF	Hard	Topo + greedy
15	Tasks Scheduling (CF 1668D)	CF	Hard	Topo DP

11.3 Shortest Paths

#	Problem	Platform	Diff	Pattern
1	Network Delay Time (743)	LC	Medium	Dijkstra
2	Cheapest Flights K Stops (787)	LC	Medium	Bellman-Ford
3	Path with Max Probability (1514)	LC	Medium	Modified Dijkstra
4	Shortest Path Binary Matrix (1091)	LC	Medium	BFS
5	Word Ladder (127)	LC	Hard	BFS + transform
6	Word Ladder II (126)	LC	Hard	BFS + backtrack
7	Swim in Rising Water (778)	LC	Hard	Dijkstra / binary search + BFS
8	Shortest Routes I	CSES	Medium	Dijkstra
9	Shortest Routes II	CSES	Medium	Floyd-Warshall
10	High Score	CSES	Hard	Bellman-Ford
11	Flight Discount	CSES	Medium	Modified Dijkstra
12	Cycle Finding	CSES	Medium	Bellman-Ford
13	Flight Routes	CSES	Hard	K-shortest paths
14	Investigation	CSES	Hard	Dijkstra + count
15	Planets Queries I	CSES	Medium	Binary lifting
16	Planets Queries II	CSES	Hard	Binary lifting

#	Problem	Platform	Diff	Pattern
17	Dijkstra? (CF 20C)	CF	Medium	Classic Dijkstra
18	Shichikuji (CF 1245D)	CF	Medium	MST/Dijkstra
19	Travelling Salesman (CF 599E)	CF	Hard	Intervals + MST
20	Mike and Fish (CF 547C)	CF	Hard	Euler + Dijkstra
21	Mahmoud and Ehab (CF 862E)	CF	Hard	Modified Dijkstra
22	Paths in a Grid (CF 1600D)	CF	Medium	Observation
23	Next Round (CF 1451D)	CF	Hard	Shortest path
24	Race (CF 196E)	CF	Hard	Dijkstra
25	Fixed Point Path (CF 1360F)	CF	Hard	BFS multi-source

11.4 DSU / MST

#	Problem	Platform	Diff	Pattern
1	Number of Provinces (547)	LC	Medium	DSU
2	Redundant Connection (684)	LC	Medium	DSU cycle
3	Accounts Merge (721)	LC	Medium	DSU
4	Satisfiability of Eq (990)	LC	Medium	DSU
5	Making a Large Island (827)	LC	Hard	DSU
6	Min Cost to Connect All (1584)	LC	Medium	Kruskal
7	Swim in Rising Water (778)	LC	Hard	Kruskal
8	Road Construction	CSES	Easy	DSU
9	Road Reparation	CSES	Medium	Kruskal
10	New Roads Queries	CSES	Hard	LCA on Kruskal tree
11	Peaceful Rooks (CF 1411C)	CF	Medium	DSU on grid
12	GCD and MST (CF 1513D)	CF	Hard	DSU + sieve
13	Consecutive Splits (CF 1629F)	CF	Hard	DSU + intervals
14	Spanning Tree Queries (CF 1033D)	CF	Hard	DSU offline
15	Reachable Nodes II (CF 1454E)	CF	Hard	DSU + topo

11.5 SCC / Bridges / Articulation Points

#	Problem	Platform	Diff	Pattern
1	Critical Connections (1192)	LC	Hard	Bridges
2	Articulation Points	GFG	Hard	Tarjan
3	Flight Routes Check	CSES	Medium	SCC check
4	Planets and Kingdoms	CSES	Hard	Kosaraju

#	Problem	Platform	Diff	Pattern
5	Giant Pizza	CSES	Hard	2-SAT
6	Coin Collector	CSES	Hard	SCC + DAG DP
7	Mail Delivery	CSES	Hard	Eulerian path
8	Teleporters Path	CSES	Hard	Eulerian path
9	De Bruijn Sequence	CSES	Hard	Eulerian circuit
10	Graph Connectivity	CSES	Hard	SCC
11	Checkposts (CF 469C)	CF	Medium	Kosaraju
12	Reachability Queries (CF 1627F)	CF	Hard	SCC + topo
13	Bridges (CF 1000E)	CF	Hard	Tarjan bridge
14	Cactus (CF 901C)	CF	Hard	SCC / cactus graph
15	2-SAT (CF 1294F)	CF	Hard	2-SAT

Chapter 12

Fenwick Tree (BIT) — Extended Problem Set

GOAL

Fenwick Tree / BIT solves range sum queries with point updates in $O(\log N)$. Master these patterns:

- **Point Update, Range Query** — Standard BIT
- **Range Update, Point Query** — Difference array BIT
- **Range Update, Range Query** — Two BITs
- **2D BIT** — For 2D range queries
- **Order Statistics** — Count elements in a range

#	Problem	Platform	Diff	Key BIT Pattern
1	Range Sum Query - Mutable (307)	LC	Medium	Standard BIT
2	Count of Smaller Numbers (315)	LC	Hard	BIT + coord compress
3	Count of Range Sum (327)	LC	Hard	Merge sort / BIT
4	Reverse Pairs (493)	LC	Hard	BIT + coord compress
5	Create Sorted Array (1649)	LC	Hard	BIT order stats
6	Number of Longest Inc Sub (673)	LC	Medium	BIT optimization
7	Russian Doll Envelopes (354)	LC	Hard	LIS via BIT
8	Max Sum of Rectangle $\leq$ K	LC	Hard	BIT on columns
9	Count Inversions	GFG	Medium	BIT + merge sort
10	Dynamic Range Sum Queries	CSES	Easy	Standard BIT
11	Range Xor Queries	CSES	Easy	BIT on XOR
12	Salary Queries	CSES	Medium	BIT + coord
13	Subarray Sum Queries	CSES	Medium	BIT prefix
14	Prefix Sum Queries	CSES	Medium	BIT
15	Range Update Queries	CSES	Medium	Difference BIT
16	Forest Queries II	CSES	Hard	2D BIT
17	Hotel Queries	CSES	Medium	BIT lower bound

#	Problem	Platform	Diff	Key BIT Pattern
18	List Removals	CSES	Medium	BIT order stats
19	Josephus Problem II	CSES	Medium	BIT kth element
20	Nested Ranges Count	CSES	Hard	BIT + coord
21	Inversion Count (CF 1093G)	CF	Hard	BIT + merge sort
22	Inverse Permutation (CF 1548C)	CF	Medium	BIT
23	Sereja and Brackets (CF 380C)	CF	Medium	Seg tree / BIT
24	Fenwick vs Segment (CF 1093G)	CF	Hard	BIT optimized
25	Lucky Numbers (CF 630F)	CF	Hard	BIT + DP
26	Little Girl and Max Sum (CF 276C)	CF	Medium	Difference BIT
27	Pashmak and Parmida (CF 459E)	CF	Hard	BIT + compress
28	Insider's Tax (CF 1611G)	CF	Hard	2D BIT
29	Numbers (CF 1285F)	CF	Hard	BIT merge
30	Array Stabilization (CF 1392F)	CF	Hard	BIT + segments
31	Long Array (CF 1893F)	CF	Hard	BIT + DP
32	Merge Sort Tree	GFG	Hard	BIT of sorted arrays
33	K-th Smallest in Range	GFG	Hard	BIT + binary search
34	Maximum Prefix Sum Queries	CF	Hard	BIT max
35	Adding Edges (CF 1691F)	CF	Hard	BIT + graph
36	Inversions After Shuffle (CF 1736E)	CF	Hard	BIT + probability
37	Vasya and a Tree (CF 1017E)	CF	Hard	DFS order + BIT
38	CGCDSSQ (CF 475D)	CF	Hard	Sparse Table / BIT
39	Range Queries and Copies (CSES)	CSES	Hard	Persistent BIT
40	Pillars (CF 660F)	CF	Hard	BIT on LIS

Chapter 13

Sparse Table — Extended Problem Set

GOAL

Sparse Table answers static range minimum/maximum queries in $O(1)$ after $O(N \log N)$ preprocessing. It is **NOT** suitable for dynamic updates (use Segment Tree / BIT for that).
Key Insight: Overlapping intervals are OK for idempotent functions (min, max, GCD, AND, OR).

When to use Sparse Table:

- Array is **static** (no updates)
- You need **many** range queries — $O(1)$ per query after build
- The function is **idempotent**: $f(f(x)) = f(x)$ (min, max, GCD, AND, OR)
- Classic use: **LCA** (via Euler tour + RMQ)

#	Problem	Platform	Diff	Key Pattern
1	Static Range Min Queries	CSES	Easy	Classic RMQ
2	Static Range Sum Queries	CSES	Easy	Prefix sum
3	Range GCD Queries	CSES	Medium	Sparse Table GCD
4	Sliding Window Maximum (239)	LC	Hard	Deque / Sparse Table
5	Longest Valid Subarray (GFG)	GFG	Medium	Sparse Table + BS
6	Maximum Sum Rectangle	GFG	Hard	Sparse Table rows
7	Minimum in Subarrays of Size K	LC	Hard	Sparse Table / deque
8	CGCDSSQ (CF 475D)	CF	Hard	Sparse Table GCD
9	Max of Min for each Window (GFG)	GFG	Hard	Sparse Table + mono
10	LCA using Euler Tour + RMQ	CSES	Hard	Sparse Table LCA
11	Distance Queries (CSES via LCA)	CSES	Medium	Euler + Sparse Table

#	Problem	Platform	Diff	Key Pattern
12	Closest Numbers (CF 459B)	CF	Easy	Sorting + RMQ-like
13	Xenia and Bit Operations (CF 339D)	CF	Medium	Seg Tree / Sparse Tbl
14	Playing with Numbers (CF 954F)	CF	Hard	Sparse Table
15	Yarik and Array (CF 1854B)	CF	Easy	Observation + sparse
16	New Year Permutation (CF 500E)	CF	Hard	RMQ-style DP
17	Range Minimum Query	SPOJ	Easy	Classic template
18	Build the Array (CF 1601B)	CF	Easy	RMQ idea
19	The Peak (CF 1768C)	CF	Medium	Sparse Table binary search
20	Two Pointer with RMQ (GFG)	GFG	Hard	Sparse Table + two-ptr
21	Longest Almost Non-Decreasing	CF	Hard	Sparse + BS
22	Subarray Minimums (GFG)	GFG	Medium	Sparse Table trick
23	Range LCM Queries (GFG)	GFG	Medium	Segment Tree (not sparse – non-idempotent)
24	RPLN (SPOJ)	SPOJ	Medium	Classic RMQ
25	RMQSQ (SPOJ)	SPOJ	Easy	Classic template
26	Ynoi2015 (2D RMQ idea)	CF	Hard	2D Sparse Table
27	CF 1093G (Inversion count)	CF	Hard	Sparse sort
28	Equal Frequencies (CF 1051G)	CF	Hard	Sparse Table + enum
29	LCA via Sparse Table	GFG	Medium	Euler tour + RMQ
30	Queries in Segtree Style	CF	Hard	Offline + Sparse Tbl

Chapter 14

Segment Tree — Extended Problem Set

GOAL

Segment Tree handles **dynamic** range queries and point updates in $O(\log N)$. **Lazy Propagation** handles **range updates** in $O(\log N)$. Master all these variants below.

#	Problem	Platform	Diff	Pattern
1	Range Sum Query - Mutable (307)	LC	Medium	Point upd, range query
2	Count of Smaller Numbers (315)	LC	Hard	Seg tree order stats
3	Count of Range Sum (327)	LC	Hard	Merge sort seg tree
4	Reverse Pairs (493)	LC	Hard	Persistent / merge
5	The Skyline Problem (218)	LC	Hard	Seg tree events
6	Dynamic Range Sum Queries	CSES	Easy	Standard seg tree
7	Dynamic Range Min Queries	CSES	Medium	Min seg tree
8	Range Xor Queries	CSES	Easy	XOR seg tree
9	Range Update Queries	CSES	Medium	Lazy seg tree
10	Hotel Queries	CSES	Medium	Seg tree lower bound
11	List Removals	CSES	Medium	Seg tree kth
12	Salary Queries	CSES	Medium	Offline + compress
13	Polynomial Queries	CSES	Hard	Lazy with formula
14	Range Updates and Sums	CSES	Hard	Lazy add + assign
15	Range Queries and Copies	CSES	Hard	Persistent seg tree
16	Distinct Values Queries	CSES	Hard	Offline + seg tree
17	Increasing Array II	CSES	Hard	Lazy seg tree
18	Sereja and Brackets (CF 380C)	CF	Medium	Interval DP on seg
19	Xenia and Bit Operations (CF 339D)	CF	Medium	Composite merge
20	Petya and Array (CF 1042D)	CF	Hard	Seg tree + DP
21	CGCDSSQ (CF 475D)	CF	Hard	Seg tree GCD

#	Problem	Platform	Diff	Pattern
22	Pashmak and Parmida (CF 459E)	CF	Hard	Coordinate compress
23	Lucky Queries (CF 1477E)	CF	Hard	Seg tree with structs
24	Vasya and a Tree (CF 1017E)	CF	Hard	DFS + seg tree
25	Ynoi Problem (CF 906E)	CF	Hard	Seg tree offline
26	Nested Ranges Count	CSES	Hard	Seg tree + coord
27	Path Queries II (CSES HLD)	CSES	Hard	HLD + seg tree
28	Distinct Colors (CSES)	CSES	Hard	DSU on tree + seg
29	Persistent Seg Tree (CF 786C)	CF	Hard	Persistent seg tree
30	Tourist's Problem (CF 1458C)	CF	Hard	Seg tree with max
31	Subarray Queries (CSES)	CSES	Medium	Seg tree
32	Recursive Seg Tree (GFG)	GFG	Medium	Build understanding
33	Multiset (CF 1354D)	CF	Medium	Seg tree multiset sim
34	Pillars (CF 660F)	CF	Hard	Seg tree on LIS
35	Segtree Beats (CF 896C)	CF	Hard	Advanced lazy

Chapter 15

String Algorithms

15.1 KMP / Z-Algorithm

#	Problem	Platform	Diff	Pattern
1	Implement strStr() (28)	LC	Easy	KMP
2	Repeated Substring Pattern (459)	LC	Easy	KMP failure func
3	Repeated String Match (686)	LC	Medium	KMP
4	Shortest Palindrome (214)	LC	Hard	KMP on reversed
5	Longest Happy Prefix (1392)	LC	Hard	Z-algorithm
6	String Matching	CSES	Easy	KMP / Z
7	Finding Borders	CSES	Medium	KMP failure func
8	Finding Periods	CSES	Medium	KMP
9	Minimal Rotation	CSES	Hard	Booth's algorithm
10	Password (CF 126B)	CF	Medium	KMP
11	Prefixes and Suffixes (CF 432D)	CF	Hard	KMP
12	Tavas and Malekas (CF 535D)	CF	Medium	KMP
13	Vasya and Beautiful Arrays (1197)	CF	Hard	Z + DP

15.2 Trie

#	Problem	Platform	Diff	Pattern
1	Implement Trie (208)	LC	Medium	Classic trie
2	Add and Search Word (211)	LC	Medium	Wildcard trie
3	Word Search II (212)	LC	Hard	Trie + DFS
4	Maximum XOR (421)	LC	Medium	Binary trie
5	Maximum XOR With Element (1707)	LC	Hard	Offline + trie
6	Concatenated Words (472)	LC	Hard	Trie + DP

#	Problem	Platform	Diff	Pattern
7	Word Combinations	CSES	Medium	Trie + DP
8	Good Substrings (CF 271D)	CF	Medium	Trie
9	Dr. Evil Underscores (CF 1285D)	CF	Hard	Trie + D&C
10	Fake News (CF 802F)	CF	Hard	Palindrome trie

15.3 Rolling Hash / Hashing

#	Problem	Platform	Diff	Pattern
1	Longest Duplicate Substring (1044)	LC	Hard	BS + rolling hash
2	Distinct Echo Substrings (1316)	LC	Hard	Rolling hash
3	Longest Chunked Palindrome (1147)	LC	Hard	Two-pointer hash
4	String Hashing	CSES	Medium	Intro to hashing
5	Longest Palindrome (CSES)	CSES	Hard	Hashing + BS
6	Repeat Count (CSES)	CSES	Hard	Hash
7	Good Substrings (CF 271D)	CF	Medium	Hash
8	Tavas and Malekas (CF 535D)	CF	Medium	Hash + KMP

Chapter 16

Massive Codeforces Problem Set

GOAL

This chapter contains **100+ curated Codeforces problems** organized by topic and difficulty. These are carefully selected from Div. 2 and Div. 3 rounds to build your pattern recognition.

16.1 CF — Div. 3 Starter (Rating 800–1300)

#	Problem	CF #	Diff	Topic
1	Way Too Long Words	71A	Easy	String basics
2	Helpful Maths	339B	Easy	Sorting + string
3	Queue at School	266B	Easy	Simulation
4	Football Kit	87A	Easy	Counting
5	Team	231A	Easy	Counting
6	Sereja and Dima	381A	Easy	Two-pointer
7	Vasya the Hipster	160A	Easy	Greedy
8	Maximize It	1649B	Easy	Observation
9	Polycarp Training	1165B	Easy	Greedy + sort
10	Array with Odd Sum	1296A	Easy	Parity
11	Queue Sorting	1561C	Easy	Observation
12	Minimize the Thickness	1838C	Easy	Greedy
13	Make It Good	1385C	Easy	Observation
14	Omkar and the Length Game	1586C	Easy	Observation
15	Three-pile Nim	768C	Easy	Game theory
16	Young Physicist	69A	Easy	Vectors
17	Numbers	1265B	Easy	XOR
18	Lucky Numbers	95A	Easy	Digit sum
19	Vanya and Lanterns	492C	Easy	Greedy
20	Next Round	334A	Easy	Counting

16.2 CF — Div. 2 Level B/C (Rating 1300–1700)

#	Problem	CF #	Diff	Topic
1	Good Subarrays	1398C	Medium	Prefix sum
2	Number of Equal Substr	1296C	Medium	Two-pointer
3	Maximum Median	1201C	Medium	BS on answer
4	Constructing the Array	1353D	Medium	Greedy + heap
5	Monsters	593B	Medium	Greedy sort
6	Balanced Rating Changes	1503B	Medium	Greedy
7	Inverse Permutation	1548C	Medium	BIT
8	Polycarp and Sums	1732C	Medium	Greedy
9	Making It Beautiful	1264C	Medium	Greedy
10	Books	279B	Medium	Sliding window
11	Salary Increasing	1550C	Medium	Greedy + BS
12	Vasya and String	1025C	Medium	Sliding window
13	Timeline	1448D	Medium	Topo sort
14	Polycarp Training	1165B	Medium	Greedy
15	Fox And Names	510C	Medium	Topo sort
16	Checkposts	469C	Medium	Kosaraju
17	Anton and Diofant	632D	Medium	GCD
18	Infinite Sequence	1216C	Medium	Observation
19	Sereja and Brackets	380C	Medium	Stack / seg tree
20	Maximum Prefix Sum	1837E	Medium	BIT
21	Letters	1187B	Medium	Binary search
22	Balanced Stone Heaps	1623C	Medium	BS + verify
23	Reachable Nodes	1234D	Medium	BFS
24	Police Stations	796D	Medium	Multi-source BFS
25	Graph Without Long Paths	1144F	Medium	BFS coloring
26	Water Tree	343D	Medium	DFS + BIT
27	Appleman and Tree	461B	Medium	Tree DP
28	Little Girl and Max Sum	276C	Medium	Diff BIT
29	Pashmak and Parmida	459E	Medium	BIT + coord
30	Peaceful Rooks	1411C	Medium	DSU
31	GCD and MST	1513D	Medium	DSU + math
32	Shichikuji and Power Grid	1245D	Medium	Dijkstra
33	Dijkstra?	20C	Medium	Dijkstra
34	Mike and Fun	547B	Medium	Row DP
35	Password	126B	Medium	KMP
36	Tavas and Malekas	535D	Medium	KMP + hash
37	Wet Shark and Flowers	621C	Medium	GCD
38	Vasya and Beautiful Arrays	1197E	Medium	Z-algo + DP
39	Good Substrings	271D	Medium	Trie / hash
40	Xenia and Bit Operations	339D	Medium	Segment tree

16.3 CF — Div. 2 Level D/E (Rating 1700–2200)

#	Problem	CF #	Diff	Topic
1	The Sports Festival	1509C	Hard	Greedy + sort
2	Devu and Flowers	451C	Hard	IE + BS
3	Increasing Subsequence	1157C2	Hard	LIS greedy
4	Maximum White Subtree	1324F	Hard	Rerooting DP
5	Lomsat Gelral	600E	Hard	DSU on tree
6	Tree Painting	1187E	Hard	Tree DP
7	Tree with Max Cost	1092F	Hard	Rerooting DP
8	Fools and Roads	191C	Hard	LCA + difference
9	Kefa and Dishes	580D	Hard	Bitmask DP
10	Counting Numbers	1202E	Hard	Digit DP
11	Lucky Queries	1477E	Hard	Seg tree struct
12	Pillars	660F	Hard	BIT on LIS
13	Petya and Array	1042D	Hard	Seg tree + DP
14	Multiset	1354D	Hard	Seg tree
15	CGCDSSQ	475D	Hard	Sparse Table GCD
16	New Year Tree	750E	Hard	HLD + seg + bits
17	Vasya and a Tree	1017E	Hard	DFS + BIT
18	Ants Colony	704D	Hard	LCA + seg tree
19	Bridges	1000E	Hard	Tarjan bridge
20	Giant Pizza	277B	Hard	2-SAT
21	Dr. Evil Underscores	1285D	Hard	Trie + D&C
22	Playing with Numbers	954F	Hard	Sparse Table
23	Spanning Tree Queries	1033D	Hard	DSU offline
24	Persistent Segment Tree	786C	Hard	Persistent seg
25	XOR Pyramid	1547F	Hard	Bitmask DP
26	Zuma Game	41D	Hard	Interval DP
27	Knapsack	1030E	Hard	Subset sum DP
28	Classy Numbers	1036C	Hard	Digit DP
29	Maximum Subset	1036D	Hard	Bitmask DP
30	Segtree Beats	896C	Hard	Advanced lazy seg
31	Tourist's Notes	57C	Hard	Bitmask DP
32	Reachability Queries	1627F	Hard	SCC + topo
33	Work for Robots	1703F	Hard	Topo + greedy
34	Adding Edges	1691F	Hard	BIT + graph
35	Inversions After Shuffle	1736E	Hard	BIT + prob
36	Long Array	1893F	Hard	BIT + DP
37	Arithmetic Progressions	1826E	Hard	DP
38	Colorful Graph	1891E	Hard	LCS-like DP
39	Equal Frequencies	1051G	Hard	Sparse Table + enum
40	Minimal String	797F	Hard	Topo + Euler path
41	Consecutive Splits	1629F	Hard	DSU + intervals
42	Reachable Nodes II	1454E	Hard	DSU + topo
43	Maximum in Segments	1870D	Hard	Greedy + BIT
44	Subsequence Counting	1748E	Hard	DP + bitmask
45	Cactus	901C	Hard	SCC cactus graph

Chapter 17

String DP

#	Problem	Platform	Diff	Pattern
1	Longest Palindromic Substr (5)	LC	Medium	Expand or DP
2	Longest Palindromic Subseq (516)	LC	Medium	2D DP
3	Palindromic Substrings (647)	LC	Medium	Expand
4	Palindrome Partitioning II (132)	LC	Hard	DP
5	Palindrome Partitioning III (1278)	LC	Hard	2D DP
6	Regular Expression Matching (10)	LC	Hard	2D DP
7	Wildcard Matching (44)	LC	Hard	2D DP
8	Word Break (139)	LC	Medium	1D DP
9	Word Break II (140)	LC	Hard	Backtrack + memo
10	Distinct Subsequences (115)	LC	Hard	2D DP
11	Interleaving String (97)	LC	Medium	2D DP
12	Scramble String (87)	LC	Hard	Memoized
13	Longest Palindrome (CSES)	CSES	Hard	Palindrome hash
14	Repeat Count (CSES)	CSES	Hard	Hashing
15	Password (CF 126B)	CF	Medium	KMP + DP

Chapter 18

Game Theory

#	Problem	Platform	Diff	Pattern
1	Nim Game (292)	LC	Easy	XOR Nim
2	Stone Game (877)	LC	Medium	Observation
3	Stone Game II (1140)	LC	Medium	DP
4	Stone Game III (1406)	LC	Hard	DP
5	Stone Game IV (1510)	LC	Hard	DP
6	Predict the Winner (486)	LC	Medium	Interval DP
7	Can I Win (464)	LC	Medium	Bitmask DP
8	Flip Game II (294)	LC	Medium	Sprague-Grundy
9	Stick Game	CSES	Easy	Grundy numbers
10	Nim Game I	CSES	Easy	XOR
11	Nim Game II	CSES	Medium	Nim variant
12	Stair Game	CSES	Medium	Nim variant
13	Grundy's Game	CSES	Hard	Grundy
14	Another Game	CSES	Hard	Grundy
15	Three-pile Nim (CF 768C)	CF	Easy	Observation
16	Petya and Strings (CF 196B)	CF	Easy	Game logic
17	Mike and Feet (CF 547B)	CF	Medium	Game DP
18	Bash and a Tough Math Puzzle	CF	Hard	Seg tree + game
19	New Year Snowflake (CF 750D)	CF	Hard	Game + tree
20	Coins Game (CF 141C)	CF	Hard	Grundy

Part III

16-Week Training Plan

Chapter 19

PHASE 1: Foundation Reinforcement (Weeks 1–4)

19.1 Week 1: Two Pointers + Sliding Window + Prefix Sum

GOAL

Master the three most fundamental array techniques. These appear in 40%+ of contest problems. **Study:** cp-algorithms "Two Pointers", CSES Sorting & Searching section.

Do problems 1–20 from Chapter 4 (Two Pointers) this week.

Daily breakdown:

- **Day 1:** Two Pointers theory + problems 1–6
- **Day 2:** Sliding Window theory + problems 7–15
- **Day 3:** Prefix Sum theory + problems 16–20
- **Day 4:** CSES problems 20–28
- **Day 5:** CF problems 29–38 (timed: 30 min each)
- **Day 6:** Live Codeforces contest
- **Day 7:** Upsolve + re-solve any hints-needed problem

19.2 Week 2: Binary Search (All Variations) + Difference Array

If you can write `check(x)` returning True/False and the answers form `[T,T,...,T,F,F,...]`, binary search on `x`.

Do problems 1–30 from Chapter 5 (Binary Search) this week.

19.3 Week 3: Bit Manipulation + Number Theory I

Do problems from Chapter 6, sections 1 (Bit Manipulation) and 2 (Number Theory) this week.

19.4 Week 4: Greedy Algorithms (Pattern-Based)

Greedy Patterns:

1. **Interval Scheduling:** Sort by end time
2. **Interval Partitioning:** Sort by start time + min-heap
3. **Fractional Knapsack:** Sort by value/weight ratio
4. **Huffman-style:** Always pick two smallest
5. **Exchange Argument:** If swapping improves, sort that way
6. **Deadline Scheduling:** Sort by deadline + priority queue

Do problems from Chapter 7 (Greedy) this week.

Chapter 20

PHASE 2: Data Structures Deep Dive (Weeks 5–8)

20.1 Week 5: Stack + Queue + Heap

Do problems from Chapter 8 (Stack, Queue, Heap) this week.

20.2 Week 6: Hashing + DSU

Study hashing patterns. Do all hashing problems from previous chapters plus DSU problems from Graphs chapter.

20.3 Week 7: Trees — DFS/BFS + Tree DP

Tree DP Patterns:
- **Pattern 1 (Subtree):** `dp[v] = f(dp[children])` (bottom-up)
- **Pattern 2 (Rerooting):** Compute for root, then propagate `dp_up[v]`
- **Pattern 3 (Choose/Skip):** `dp[v][0]` = not chosen, `dp[v][1]` = chosen

Do problems from Chapter 9 (Trees) this week.

20.4 Week 8: Fenwick Tree (BIT) + Sparse Table + LCA

This is a dedicated week for Chapter 10 (BIT) and Chapter 11 (Sparse Table) problems.

Common BIT Mistakes:
- Forgetting 1-indexed arrays
- Wrong direction of propagation in range-update BIT
- Off-by-one in coordinate compression

Chapter 21

PHASE 3: Graphs + DP Mastery (Weeks 9–12)

21.1 Week 9: Graphs — BFS/DFS + Bipartite + Topological Sort

Do all BFS/DFS and Topological Sort problems from Chapter 12.

Algorithm Selection Guide:

Problem Type	Algorithm
Unweighted graph shortest path	BFS
Non-negative weights	Dijkstra $O((V+E)\log V)$
Negative weights, no neg cycle	Bellman-Ford $O(VE)$
All pairs, small V	Floyd-Warshall $O(V^3)$
MST	Kruskal (sort + DSU) or Prim
Bridges / Articulation Points	Tarjan's DFS
SCC	Kosaraju's or Tarjan's
Cycle in directed graph	DFS coloring

21.2 Week 10: Shortest Paths + MST + SCC

Do all shortest path, DSU/MST, and SCC problems from Chapter 12.

21.3 Week 11: DP Mastery I (1D, 2D, Knapsack, LIS, LCS)

THIS IS THE MOST IMPORTANT WEEK. DP IS KING.
Do ALL problems from Chapter 8 sections 1–5 (1D DP, 2D DP, Knapsack, LIS/LCS, String DP).

21.4 Week 12: DP Mastery II (Bitmask, Digit, Interval, Tree DP)

Do all remaining DP problems: Interval DP, Bitmask DP, Digit DP, Tree DP from Chapter 8.

Chapter 22

PHASE 4: Advanced Topics + Contest Integration (Weeks 13–16)

22.1 Week 13: Segment Tree + Fenwick Tree + Sparse Table

Do ALL problems from Chapters 10, 11, 12 (BIT Extended, Sparse Table Extended, Segment Tree Extended). This is a full week dedicated to advanced range query data structures.

22.2 Week 14: String Algorithms

Do all problems from Chapter 13 (Strings): KMP, Z-algorithm, Trie, Rolling Hash, String DP.

22.3 Week 15: Math + Combinatorics + Game Theory

Game Theory Core:
Nim: XOR of all pile sizes. XOR = 0 → second player wins.
Sprague-Grundy: Every impartial game position has a Grundy number.
Grundy(pos) = mex({Grundy(next positions)})
Combined game: XOR of Grundy numbers of sub-games.

Do all math, combinatorics, and game theory problems.

22.4 Week 16: Integration + Mixed Practice + Contest Simulation

NO NEW TOPICS — PURE PRACTICE AND REVISION

- Day 1–2: Virtual Contest (CF Div 2 from 2023–2024) + Upsolve
- Day 3: Solve 5 problems from your mistake log
- Day 4–5: Another Virtual Contest + Upsolve
- Day 6: Live Contest
- Day 7: Final review of hardest problems

Part IV

Top Lists & Revision Strategy

Chapter 23

Top 50 Must-Do Problems

#	Problem	Platform	Key Pattern
Arrays / Two Pointers / Sliding Window			
1	3Sum (LC 15)	LC	Two Pointers + Sort
2	Container With Most Water (11)	LC	Two Pointers
3	Minimum Window Substring (76)	LC	Sliding Window
4	Subarrays K Different Int (992)	LC	At Most K Trick
5	Trapping Rain Water (42)	LC	Stack / Two Pointer
Binary Search			
6	Koko Eating Bananas (875)	LC	BS on Answer
7	Split Array Largest Sum (410)	LC	BS on Answer
8	Median of Two Sorted Arrays (4)	LC	BS Partitioning
9	Factory Machines	CSES	BS on Answer
Stack / Queue / Heap			
10	Largest Rectangle Histogram (84)	LC	Monotonic Stack
11	Sliding Window Maximum (239)	LC	Monotonic Deque
12	Find Median Data Stream (295)	LC	Two Heaps
13	Sum Subarray Minimums (907)	LC	Stack + Contribution
Trees			
14	BT Max Path Sum (124)	LC	Tree DFS
15	Serialize/Deserialize BT (297)	LC	Design
16	Construct from Pre/In (105)	LC	Divide & Conquer
17	Sum Distances in Tree (834)	LC	Rerooting
18	Tree Distances II (CSES)	CSES	Rerooting DP
Graphs			
19	Course Schedule II (210)	LC	Topological Sort

#	Problem	Platform	Key Pattern
20	Word Ladder (127)	LC	BFS
21	Network Delay Time (743)	LC	Dijkstra
22	Critical Connections (1192)	LC	Bridges
23	Shortest Routes I (CSES)	CSES	Dijkstra
24	Planets and Kingdoms (CSES)	CSES	SCC
Dynamic Programming			
25	Coin Change (322)	LC	Unbounded Knapsack
26	Partition Equal Subset (416)	LC	0/1 Knapsack
27	LIS (300)	LC	$O(N \log N)$ LIS
28	Edit Distance (72)	LC	2D DP
29	Word Break (139)	LC	1D DP + HashSet
30	Best Time Buy/Sell III (123)	LC	State Machine DP
31	Regex Matching (10)	LC	2D DP
32	Burst Balloons (312)	LC	Interval DP
33	LCS (1143)	LC	2D DP
34	Hamiltonian Flights (CSES)	CSES	Bitmask DP
35	Counting Numbers (CSES)	CSES	Digit DP
36	Russian Doll Envelopes (354)	LC	LIS in 2D
37	Rod Cutting (GFG)	GFG	Unbounded Knapsack
38	Matrix Chain Multiplication (GFG)	GFG	Interval DP
BIT / Sparse Table / Seg Tree			
39	Count of Smaller Numbers (315)	LC	BIT + Coord Compress
40	Range Sum Mutable (307)	LC	BIT / Seg Tree
41	Distinct Values Queries (CSES)	CSES	Offline + Seg Tree
42	CGCDSSQ (CF 475D)	CF	Sparse Table GCD
43	Salary Queries (CSES)	CSES	BIT + Compress
Strings			
44	Longest Palindromic Substr (5)	LC	DP / Expand
45	Word Search II (212)	LC	Trie + DFS
46	Shortest Palindrome (214)	LC	KMP
47	Longest Happy Prefix (1392)	LC	Z-algorithm
Math / Bit / Mixed			
48	Single Number III (260)	LC	XOR + Bit Split
49	Swim in Rising Water (778)	LC	BS + BFS
50	Longest Inc. Path Matrix (329)	LC	DFS + Memo

Chapter 24

Last 2-Week Revision Strategy

24.1 Revision Week 1: Pattern Recall

Day	Activity
1	Re-solve: LC 76, 875, 410, 4 (each $<$ 15 min)
2	Re-solve: LC 84, 239, 295 (each $<$ 20 min)
3	Re-solve: LC 124, 297, 210, CSES Shortest Routes I
4	Re-solve: LC 322, 416, 300, 72, 1143 (10–15 min each)
5	Re-solve: LC 312, 10, 354, CSES Hamiltonian Flights
6	Full virtual CF Div 2 contest
7	Catch-up: re-solve anything slow + review mistake log

24.2 Revision Week 2: Speed + Accuracy

Day	Activity
1–2	20 random LC Mediums in 2 hrs (6 min each target)
3	Mock interview (Pramp / friend): 2 problems in 45 min
4–5	Final virtual CF Div 2 (full 2 hrs) + upsolve
6	Review ALL code templates (BS, DSU, SegTree, BIT, Sparse Table, LCA)
7	Rest + 2–3 easy problems + final mistake log review

Chapter 25

Common Mistakes & How to Avoid Them

15 MISTAKES THAT COST YOU RATINGS

1. **Not reading constraints** — FIX: Look at N first, write expected complexity.
2. **Jumping to code too fast** — FIX: 5-minute rule, pen and paper first.
3. **Not drawing examples** — FIX: ALWAYS draw 3 examples before coding.
4. **Not testing edge cases** — FIX: Always test $N = 1$, all same, max constraints.
5. **Integer overflow** — FIX: `long long` by default. Any $n \times n > 10^4$ must be `long long`.
6. **Wrong binary search boundaries** — FIX: Use consistent template every time.
7. **Forgetting to reset between test cases** — FIX: ALL init inside `solve()`.
8. **TLE due to unnecessary operations** — FIX: Avoid `endl`, pass by reference.
9. **Greedy without proof** — FIX: Try 3 small counterexamples. Use exchange argument.
10. **DP state too large** — FIX: Rolling array, state compression, meet in the middle.
11. **Not upsolving** — FIX: Do not participate in next contest until you upsolve the previous one.
12. **Off-by-one in BIT** — FIX: Always use 1-indexed BIT. Add offset for 0-indexed input.
13. **Wrong idempotent function in Sparse Table** — FIX: Only use for min/-max/GCD/AND/OR.
14. **Confusing BIT vs Seg Tree** — FIX: BIT for prefix sums, Seg Tree for range queries with lazy.
15. **Panic in contests** — FIX: Read ALL problems first. Solve the easiest you know. Never spend > 30 min on a problem without getting something.

Part V

DevOps & Cloud Engineering — 8 Weeks

Chapter 26

DevOps Roadmap Summary

26.1 Week-by-Week Overview

Week	Topic	Key Tools	Project
1	Docker Fundamentals	Docker, Dockerfile	Multi-stage optimization
2	Docker Networking	Docker Compose, Volumes	Full-stack Docker app
3	CI/CD with GitHub Actions	YAML, Runners, Secrets	Auto deployment pipeline
4	Kubernetes Foundations	Minikube, kubectl, Pods	Deploy Nginx to K8s
5	K8s Networking + Helm	Ingress, PVC, Helm Charts	Full-stack K8s deploy
6	Terraform (IaC)	Terraform, AWS/GCP	Provision VPC + VM
7	Monitoring + Observability	Prometheus, Grafana, ELK	Custom dashboard
8	DevSecOps	Trivy, Vault, RBAC	Secure GitOps deploy

26.2 Optimized Multi-Stage Dockerfile

```
# Stage 1: Build
FROM node:18-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm ci --only=production
COPY . .
RUN npm run build

# Stage 2: Production
FROM nginx:stable-alpine
COPY --from=builder /app/dist /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon␣off;"]
```

Part VI

Summary Statistics

Chapter 27

Total Problem Count

Topic	LC	CF	CSES/Other
Two Pointers / Sliding Window	25	13	9
Binary Search	23	12	10
Bit Manipulation	20	10	0
Number Theory & Math	12	10	18
Greedy	18	15	12
Stack / Queue / Heap	30	10	5
Dynamic Programming	60	20	22
Trees	30	10	20
Graphs	35	30	25
BIT (Extended)	8	22	10
Sparse Table (Extended)	5	15	10
Segment Tree (Extended)	5	20	10
Strings	20	15	10
Massive CF Problem Set	0	85	0
String DP	13	2	0
Game Theory	8	7	5
TOTAL	**312**	**296**	**166**
Grand Total			**≈774**

How to reach 1600+ on Codeforces from your current 1551:

1. Solve 5 problems per day from this list
2. Participate in every live Codeforces round (Div. 2)
3. Upsolve the next unsolved problem after every contest
4. After 3 months of this, you will be comfortably solving Div. 2 D problems
5. After 6 months, you will be targeting Expert (1600+)

Remember: You already have 1551 rating and 400+ LC problems solved. You are **closer than you think!**